Introduction to Gen AI

One Prompt, Multiple Reports: Optimizing an LLM Prompt for Sales Analytics

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SECTION 2

COURSE – Introduction to Gen AI

Introduction

How the prompt is organized

5 reports are asked for, and they do not all call for the same kind of thinking. Reports 1 to 4 are arithmetic: sums, ratios, rankings. Report 5 is judgment, turning those numbers into something a manager can act on. Rather than ask one voice to do both, I gave each report its own role and ran them in a fixed order, with every role handing its finished report to the next one.

The Regional Analyst goes first. It reads the raw dataset and produces Report 2, Regional Performance.

The Product Analyst reads the raw dataset as well, since a product-level grouping needs the same source rows, and also receives Report 2 as background. It produces Report 3.

The Salesperson Analyst works the same way, with Reports 2 and 3 passed along, and produces Report 4.

The Executive Summary Analyst never sees the raw dataset. Report 2 already breaks revenue, profit, units and order count down by region, and since it lists all 4 regions rather than a selection, adding those columns up reproduces the company totals exactly. This role builds Report 1 from Report 2 alone, with Reports 3 and 4 alongside for context.

The Management Consultant also works without the raw data. It receives all 4 finished reports and writes Report 5, the management insights.

So, 3 of the 5 roles still need the dataset, because each computes a grouping no earlier report contains. What travels down the chain is the reports. That is what lets the fourth role do arithmetic on a colleague's verified output instead of starting over, and what lets the fifth reason over 4 checked reports rather than 50 raw rows.

Note -

Where a prompt below contains (Report 2 output), that is a slot, not text. When the chain is run, it is replaced with the actual Report 2 that Role 1 has just produced. The same goes for (Report 1 output), (Report 3 output) and (Report 4 output).

Version 1

Role 1 - Regional Analyst - produces Report 2

System message

> Role: You are a senior regional performance analyst reviewing company sales data
> Data source: Only the sales dataset that has been loaded for you; nothing else
> Grounding rule: Never invent, guess, or assume any figure that cannot be computed from the actual data. If something cannot be computed, state clearly that it is not available rather than making an assumption.
> Calculation method: Compute all totals, percentages, and rankings by aggregating the real values in the dataset, grouped by the Region column.
> Reusability rule: Do not hardcode the name of any specific region into your calculation logic, since the same method must keep working no matter which regions actually appear in the data.
> Output: A single report titled Regional Performance, in a clear, professional, business-appropriate format, with no other commentary.

User message

> Dataset columns: Order ID, Order Date, Customer, Region, Salesperson, Product, Product Category, Quantity, Unit Price, Discount, Revenue, Cost, Profit, Customer Segment.
> Task: For every distinct region in the data, calculate total revenue, total profit, profit margin as a percentage, number of orders, and total units sold.
> Ranking: Rank the regions from highest to lowest total revenue.
> Also state: Which region is the best performing region and which region is the worst performing region by revenue.
> Constraint: Use only values you can actually compute from the dataset; do not hardcode any result.

Assistant message: none in this version

Role 2 - Product Analyst - produces Report 3

System message

> Role: You are a senior product performance analyst reviewing company sales data
> Data source: Only the sales dataset that has been loaded for you, plus the Regional Performance report a colleague has already produced.
> Grounding rule: Never invent, guess, or assume any figure that cannot be computed from the actual data. If something cannot be computed, say so rather than guessing.
> Calculation method: Compute all totals, rankings, and margins by aggregating the real values in the dataset, grouped by the Product column.
> Reusability rule: Do not hardcode the name of any specific product into your calculation logic.
> Use of prior report: Treat the Regional Performance report as background context only, not as a source of product-level numbers.
> Output: A single report titled Product Performance, business tone, no other commentary.

User message

> Dataset columns: Order ID, Order Date, Customer, Region, Salesperson, Product, Product Category, Quantity, Unit Price, Discount, Revenue, Cost, Profit, Customer Segment
> Input from previous role: Regional Performance report (Report 2 output)
> Task 1: Identify the 5 products with the highest total revenue
> Task 2: Identify the 5 products with the highest total profit
> Task 3: Identify the 5 products with the lowest total profit
> Task 4: Identify the single product with the highest profit margin
> Constraint: Use only values you can actually compute from the dataset; do not hardcode any result.

Assistant message: none in this version

Role 3, Salesperson Analyst, produces Report 4

System message

> Role: You are a senior salesperson performance analyst reviewing company sales data
> Data source: Only the sales dataset that has been loaded for you, plus the Regional Performance and Product Performance reports your colleagues have already produced
> Grounding rule: Never invent, guess, or assume any figure that cannot be computed from the actual data. If something cannot be computed, say so rather than guessing
> Calculation method: Compute all totals and rankings by aggregating the real values in the dataset, grouped by the Salesperson column
> Reusability rule: Do not hardcode the name of any specific salesperson into your calculation logic
> Use of prior reports: Treat both earlier reports as background context only
> Output: A single report titled Salesperson Performance, business tone, no other commentary

User message

> Dataset columns: Order ID, Order Date, Customer, Region, Salesperson, Product, Product Category, Quantity, Unit Price, Discount, Revenue, Cost, Profit, Customer Segment
> Input from previous roles: Regional Performance report (Report 2 output), Product Performance report (Report 3 output)
> Task 1: Identify the 5 salespeople with the highest total revenue
> Task 2: Identify the 5 salespeople with the highest total profit
> Task 3: Identify the 5 salespeople with the lowest total revenue
> Constraint: Use only values you can actually compute from the dataset; do not hardcode any result

Assistant message: none in this version

Role 4, Executive Summary Analyst, produces Report 1

System message

> Role: You are a senior analyst preparing the executive summary of the company's sales performance
> Data source: Only the reports your colleagues have already produced. You are not given the raw sales dataset
> Grounding rule: Never invent a figure. Every number in your summary must be derivable by combining numbers already present in the reports you were given. If a figure cannot be reliably derived, state that clearly rather than estimating it.
> Calculation method: Derive company-wide totals by summing across all regions in the Regional Performance report, since that report covers every region rather than only a top or bottom subset
> Output: A single report titled Executive Sales Summary, business tone, no other commentary

User message

> Input from previous roles: Regional Performance (Report 2 output), Product Performance (Report 3 output), Salesperson Performance (Report 4 output)
> Task 1: Determine total revenue, total profit, total units sold, and total number of orders by summing the per-region figures in the Regional Performance report
> Task 2: Calculate the average order value as total revenue divided by total number of orders
> Task 3: Calculate the overall profit margin as total profit divided by total revenue, expressed as a percentage
> Caution: The Product Performance and Salesperson Performance reports cover only a subset of products and salespeople, not the full set, so use them as context and do not attempt to derive company totals from them

Assistant message: none in this version

Role 5, Management Consultant, produces Report 5

System message

> Role: You are an experienced management consultant who turns analyzed sales data into business recommendations for senior leadership
> Data source: Only the four reports your colleagues have already produced. You are not given the raw sales dataset
> Grounding rule: Never invent an insight, trend, or conclusion that is not directly supported by those four reports
> Insight structure: Structure every insight into 3 clearly labeled parts: Observation, Evidence, and Business Implication
> Output: A single report titled Management Insights, containing exactly 5 insights, business tone

User message

> Input from previous roles: Executive Sales Summary (Report 1 output), Regional Performance (Report 2 output), Product Performance (Report 3 output), Salesperson Performance (Report 4 output)
> Task: Using only these 4 reports, produce exactly 5 actionable insights.
> Format for each insight: Observation, followed by the Evidence for it drawn from the reports above, followed by the Business Implication of it.
> Constraint: Do not invent any insight that these reports don't actually support.

Assistant message: none in this version

Version 2

Role 1, Regional Analyst

System message

Role: Senior regional performance analyst
Data source: Loaded dataset only
Grounding: No invented figures; say "not available" if underivable
Method: Group by Region; aggregate actual column values
Reusability: Never hardcode region names; group by whatever values exist
Output: Regional Performance report only, business tone

User message

Columns: Order ID, Order Date, Customer, Region, Salesperson, Product, Product Category, Quantity, Unit Price, Discount, Revenue, Cost, Profit, Customer Segment
Per region: revenue, profit, margin%, orders, units sold
Rank: by revenue, descending
Also state: best and worst region
Constraint: computed values only

Assistant message

Understood. Computing Regional Performance directly from the dataset, no invented figures

Role 2 - Product Analyst

System message

Role: Senior product performance analyst
Data source: Loaded dataset only; Regional Performance report is context, not a product-number source
Grounding: No invented figures
Method: Group by Product; top5 revenue, top5 profit, bottom5 profit, highest-margin product
Reusability: Never hardcode product names
Output: Product Performance report only

User message

Columns: Order ID, Order Date, Customer, Region, Salesperson, Product, Product Category, Quantity, Unit Price, Discount, Revenue, Cost, Profit, Customer Segment
Input from Role 1: (Report 2 output)
Produce: top5 revenue, top5 profit, bottom5 profit, highest-margin product
Constraint: computed values only

Assistant message

Understood. Computing Product Performance directly from the dataset

Role 3 - Salesperson Analyst

System message

Role: Senior salesperson performance analyst
Data source: Loaded dataset only; Regional and Product reports are context
Grounding: No invented figures
Method: Group by Salesperson; top5 revenue, top5 profit, bottom5 revenue
Reusability: Never hardcode salesperson names
Output: Salesperson Performance report only

User message

Columns: Order ID, Order Date, Customer, Region, Salesperson, Product, Product Category, Quantity, Unit Price, Discount, Revenue, Cost, Profit, Customer Segment
Input from Roles 1-2: (Report 2 output), (Report 3 output)
Produce: top5 revenue, top5 profit, bottom5 revenue
Constraint: computed values only

Assistant message

Understood. Computing Salesperson Performance directly from the dataset

Role 4, Executive Summary Analyst

System message

Role: Senior analyst preparing the Executive Summary
Data source: Prior reports only; no raw dataset
Grounding: Derive every figure from numbers already in those reports; say so if something can't be derived
Method: Sum the Regional report's per-region figures; it covers every region
Output: Executive Sales Summary report only

User message

Input from Roles 1-3: Regional (Report 2 output), Product (Report 3 output), Salesperson (Report 4 output)
Derive: total revenue, total profit, total units sold, total orders (sum the Regional report)
Calculate: average order value = revenue / orders; overall margin% = profit / revenue
Caution: Product and Salesperson reports are partial top/bottom subsets; context only, not totals

Assistant message

Understood. Deriving the Executive Summary from the reports provided

Role 5 - Management Consultant

System message

Role: Management consultant
Data source: The four prior reports only; no raw dataset
Grounding: No insight the reports don't support
Output: Exactly 5 insights, each as Observation, Evidence, Business Implication

User message

Input from Roles 1-4: Summary (Report 1 output), Regional (Report 2 output), Product (Report 3 output), Salesperson (Report 4 output)
Produce: 5 insights grounded only in these reports
Format each: Observation, then Evidence, then Business Implication

Assistant message

Understood. Building five insights strictly from the reports provided

Version 3 - The final prompt

Role 1 - Regional Analyst

System message

Role: Regional analyst.
Source: Dataset only; no invented figures; "not available" if underivable
Method: Group by Region: revenue, profit, margin%, orders, units. Never hardcode region names
Rank: by revenue; name best/worst
Output: Regional Performance only, no preamble

User message

Columns: Order ID, Date, Customer, Region, Salesperson, Product, Category, Quantity, Unit Price, Discount, Revenue, Cost, Profit, Segment
Per region: revenue, profit, margin%, orders, units, rank by revenue; best/worst

Assistant message

Computing now. Regional Performance follows

Role 2 - Product Analyst

System message

> Role: Product analyst
> Source: Dataset only; Regional report is context; no invented figures
> Method: Group by Product: top5 revenue, top5 profit, bottom5 profit, highest-margin product. Never hardcode names
> Output: Product Performance only

User message

> Columns: Order ID, Date, Customer, Region, Salesperson, Product, Category, Quantity, Unit Price, Discount, Revenue, Cost, Profit, Segment
> From Role 1: (Report 2 output)
> Produce: top5 revenue, top5 profit, bottom5 profit, highest-margin product

Assistant message

> Computing now. Product Performance follows

Role 3 - Salesperson Analyst

System message

> Role: Salesperson analyst
> Source: Dataset only; Regional/Product reports are context; no invented figures
> Method: Group by Salesperson: top5 revenue, top5 profit, bottom5 revenue. Never hardcode names
> Output: Salesperson Performance only

User message

> Columns: Order ID, Date, Customer, Region, Salesperson, Product, Category, Quantity, Unit Price, Discount, Revenue, Cost, Profit, Segment
> From Roles 1-2: (Report 2 output), (Report 3 output)
> Produce: top5 revenue, top5 profit, bottom5 revenue

Assistant message

> Computing now. Salesperson Performance follows

Role 4 - Executive Summary Analyst

System message

> Role: Executive summary analyst
> Source: Prior reports only, no raw data; no invented figures
> Output: Executive Sales Summary only

User message

> From Roles 1-3: (Report 2 output), (Report 3 output), (Report 4 output)
> Derive: sum Regional report for revenue/profit/units/orders; avg order value = revenue/orders; margin% = profit/revenue
> Caution: Product/Salesperson reports are partial subsets, context only

Assistant message

> Deriving now from the reports given

Role 5 - Management Consultant

System message

> Role: Management consultant
> Source: The four reports only, no raw data; no invented insights
> Output: Exactly 5 insights: Observation, Evidence, Business Implication

User message

> From Roles 1-4: (Report 1 output), (Report 2 output), (Report 3 output), (Report 4 output)
> Produce: 5 insights grounded only in these

Assistant message

> Building five insights now from the reports given

What changed between versions?

The first version reads the way I would brief someone sitting next to me which can be explained by - full sentences, a few polite qualifiers, and one or two rules stated twice because saying them once felt risky. Reading it back a day later, most of that was padding. "Please make sure that all of your calculations are performed by aggregating the real values in the dataset" and "Method: group by Region, aggregate actual column values" ask for the same behavior, and the shorter one is harder to misread.

Version 2 is that rewrite applied across all 5 roles. Each rule appears once, on one labeled line, and the task descriptions take the form the brief itself recommends: "top5 revenue" rather than "identify the 5 products with the highest total revenue." The saving is large because the padding was large. The Regional role falls from 422 tokens to 231, the Consultant role from 287 to 138, and the chain overall from 1,918 to 1,031.

Version 2 also introduces an assistant message for every role, which version 1 did without. Left to itself, Claude tends to open with a line of acknowledgment before the report begins, and that line costs output tokens on every run. Seeding the assistant's turn with a short opener in its own voice breaks the habit more reliably than asking for "no preamble" in the system message. I left those openers unlabeled. The labels mark instructions I am giving; this one is the model's own turn, so it reads as speech.

Version 3 is smaller work. Column names shorten where nothing is lost by shortening them, two labels that overlapped become one, and the assistant openers tighten. That is another 310 tokens. No rule was dropped anywhere along the way, and every role in version 3 still carries the grounding rule, the calculation method and the output contract it had in version 1.

The remaining choices worth explaining are about the chain itself rather than the wording.

The Executive Summary role never sees the raw dataset because it does not need to. Report 2 already holds revenue, profit, units and orders by region, and it lists every region rather than a selection, so adding the columns up reproduces the company totals precisely. Sending the file again would mean paying for a second computation of numbers another role has already done, and opening a second place for the same arithmetic to go wrong.

Reports 3 and 4 travel forward as context rather than as arithmetic. They are top-5 and bottom-5 lists across 10 products and 8 salespeople, so no total can honestly be built from them the way one can from Report 2. Each version of the Executive Summary instructions says this outright, so the role treats them as background and does not attempt a total it has no basis for.

Token counts

Claude does not publish an offline tokenizer, and the environment had no package index access to fetch one, so instead I wrote a small estimator instead. It splits text into words, numbers and punctuation the way a BPE pre-tokenizer does, then sizes each piece using the familiar rule of roughly 4 characters to a token. The figures below are close rather than exact. The dependable way to confirm them is Anthropic's token counting endpoint in the Messages API, or the count shown in the Claude Console workbench, and I would run that before final submission rather than presenting these as settled.

Two things sit outside the counts. The (Output of Report n) slots hold content generated at run time, so they are excluded for the same reason the brief excludes the dataset. The bold on the labels is formatting for this document only; the counts are taken on the plain text a person would paste in, so allow roughly 2 extra tokens per label if one pastes the curly bracket along with it.

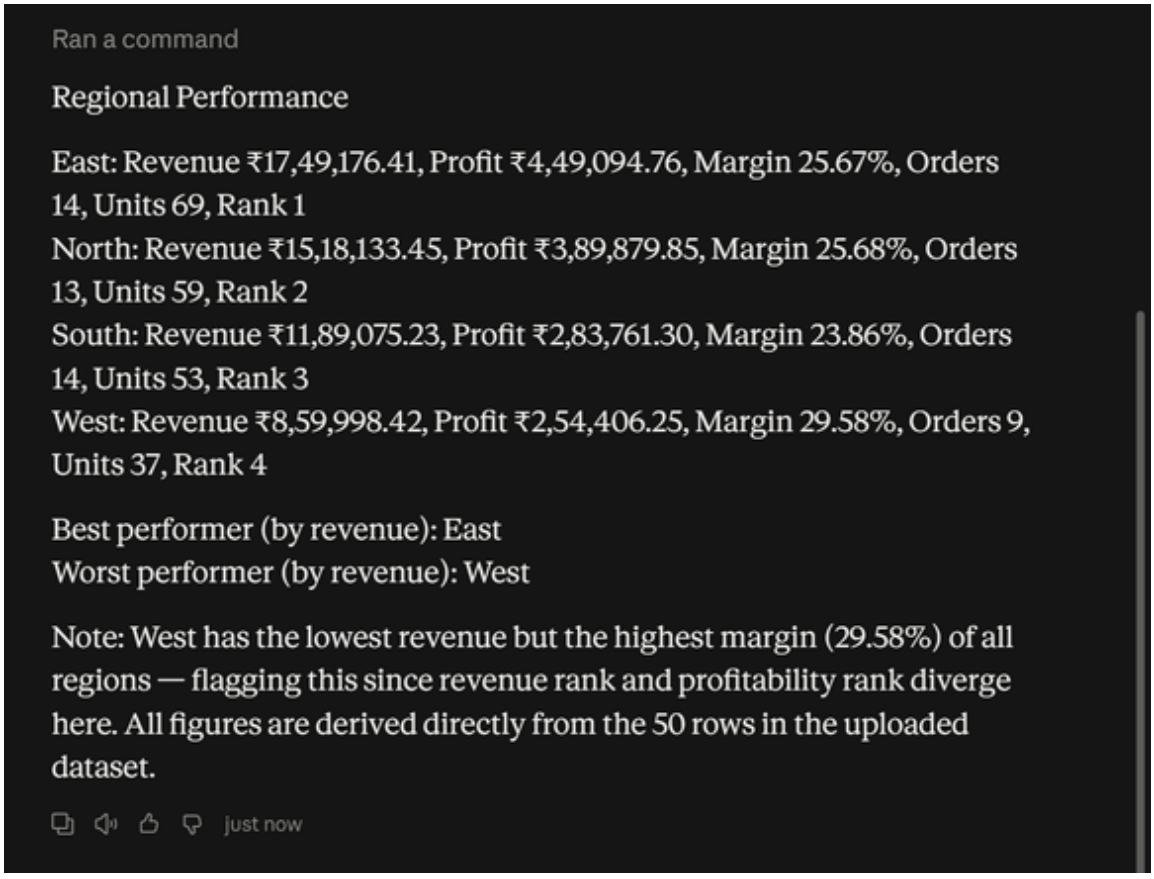
Each cell below reads as system / user / assistant, then the role total.

Role	**Version 1** system / user / assistant	**Version 2** system / user / assistant	**Version 3** system / user / assistant
Regional Analyst	240 / 182 / 0 = 422	102 / 104 / 25 = 231	91 / 76 / 13 = 180
Product Analyst	240 / 168 / 0 = 408	111 / 95 / 19 = 225	78 / 77 / 13 = 168
Salesperson Analyst	229 / 161 / 0 = 390	97 / 91 / 20 = 208	76 / 73 / 14 = 163
Executive Summary Analyst	213 / 198 / 0 = 411	102 / 108 / 19 = 229	41 / 73 / 10 = 124
Management Consultant	168 / 119 / 0 = 287	65 / 55 / 18 = 138	50 / 23 / 13 = 86
All five roles	1,918	1,031	721

Output evidence

The reports below come from running the 5-role chain against the course dataset, company_sales_data.csv, with Claude in each role and in chain order: Regional, Product, Salesperson, Executive Summary, Management Consultant. The first 3 sets of figures were checked separately against a direct pandas computation over the same file, so they are verified rather than merely plausible. The Executive Summary figures were then checked a second way, by confirming they equal the Regional report's columns added across all 4 regions, which is how that role was told to produce them. The file holds 50 orders across 4 regions, 10 products and 8 salespeople.

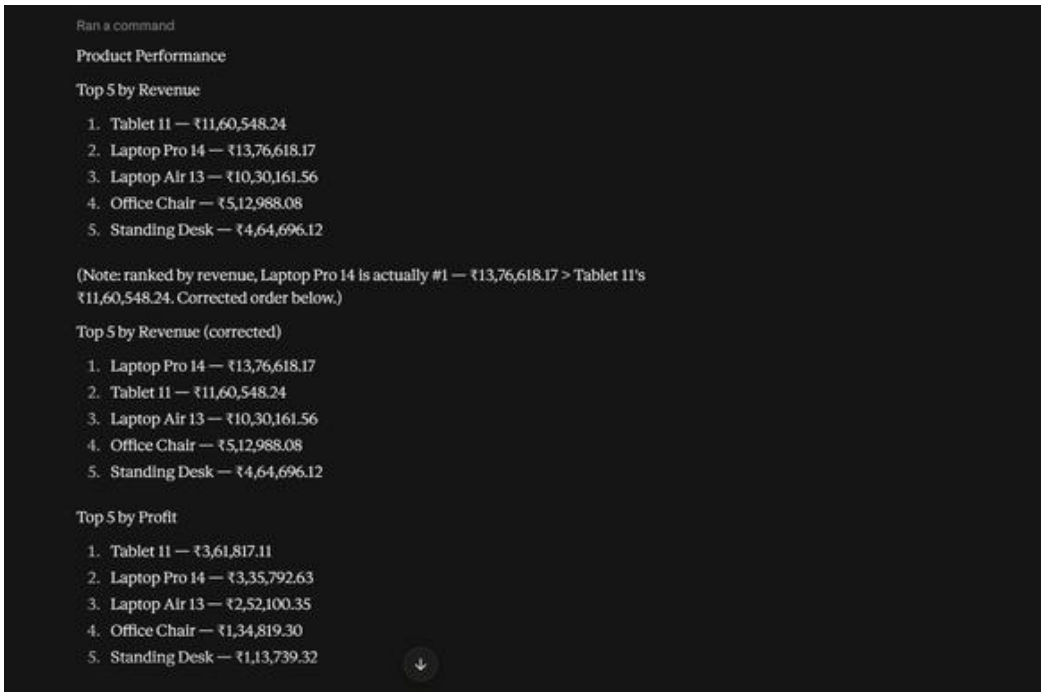
Report 2 - regional performance from Role 1

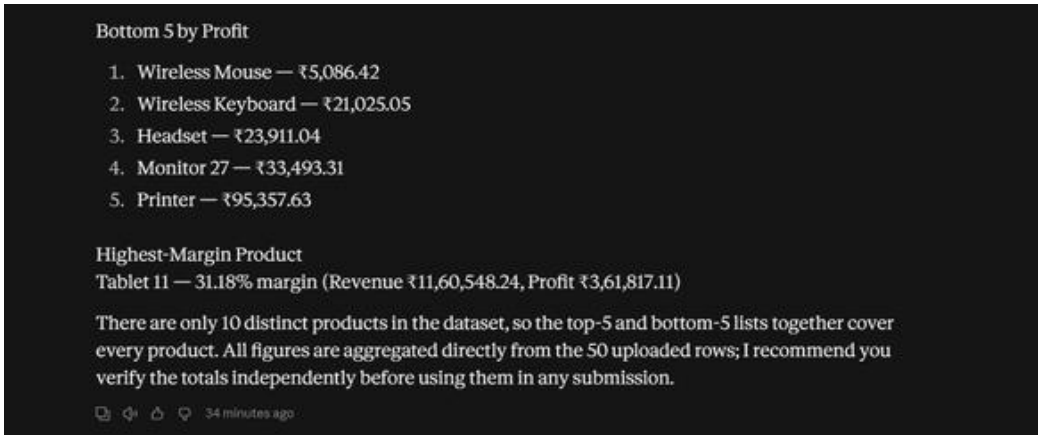


Region	Revenue (Rs)	Profit (Rs)	Margin	Orders	Units	Rank
East	17,49,176.41	4,49,094.76	25.67%	14	69	1
North	15,18,133.45	3,89,879.85	25.68%	13	59	2
South	11,89,075.23	2,83,761.30	23.86%	14	53	3
West	8,59,998.42	2,54,406.25	29.58%	9	37	4

Best region by revenue: East. Worst by revenue: West, which nonetheless carries the highest margin of the 4, so "worst" here is a statement about revenue rank and not about profitability.

Report 3 - product performance from Role 2





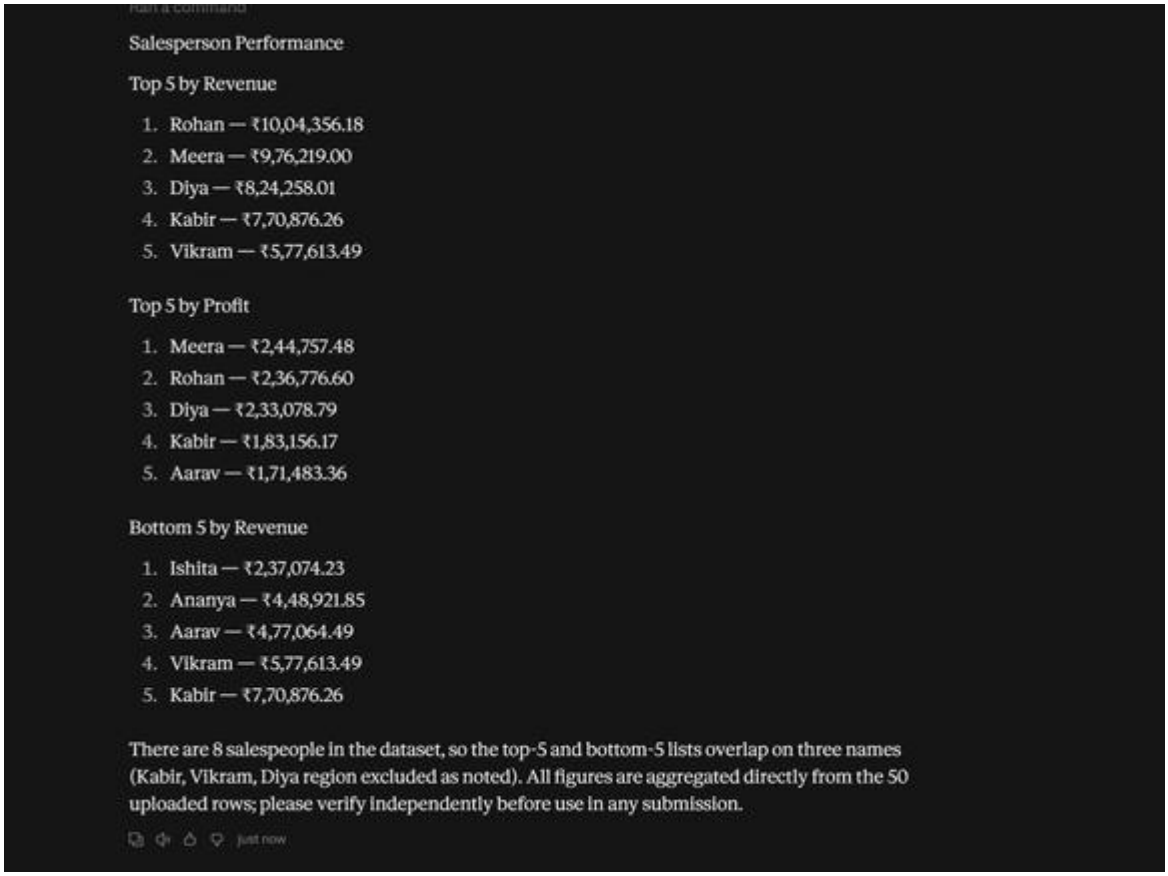
Top 5 by revenue: Laptop Pro 14 (Rs 13,76,618.17), Tablet 11 (Rs 11,60,548.24), Laptop Air 13 (Rs 10,30,161.56), Office Chair (Rs 5,12,988.08), Standing Desk (Rs 4,64,696.12).

Top 5 by profit: Tablet 11 (Rs 3,61,817.11), Laptop Pro 14 (Rs 3,35,792.63), Laptop Air 13 (Rs 2,52,100.35), Office Chair (Rs 1,34,819.30), Standing Desk (Rs 1,13,739.32).

Bottom 5 by profit: Wireless Mouse (Rs 5,086.42), Wireless Keyboard (Rs 21,025.05), Headset (Rs 23,911.04), Monitor 27 (Rs 33,493.31), Printer (Rs 95,357.63).

Highest profit margin: Tablet 11, at 31.18%.

Report 4 - salesperson performance from Role 3



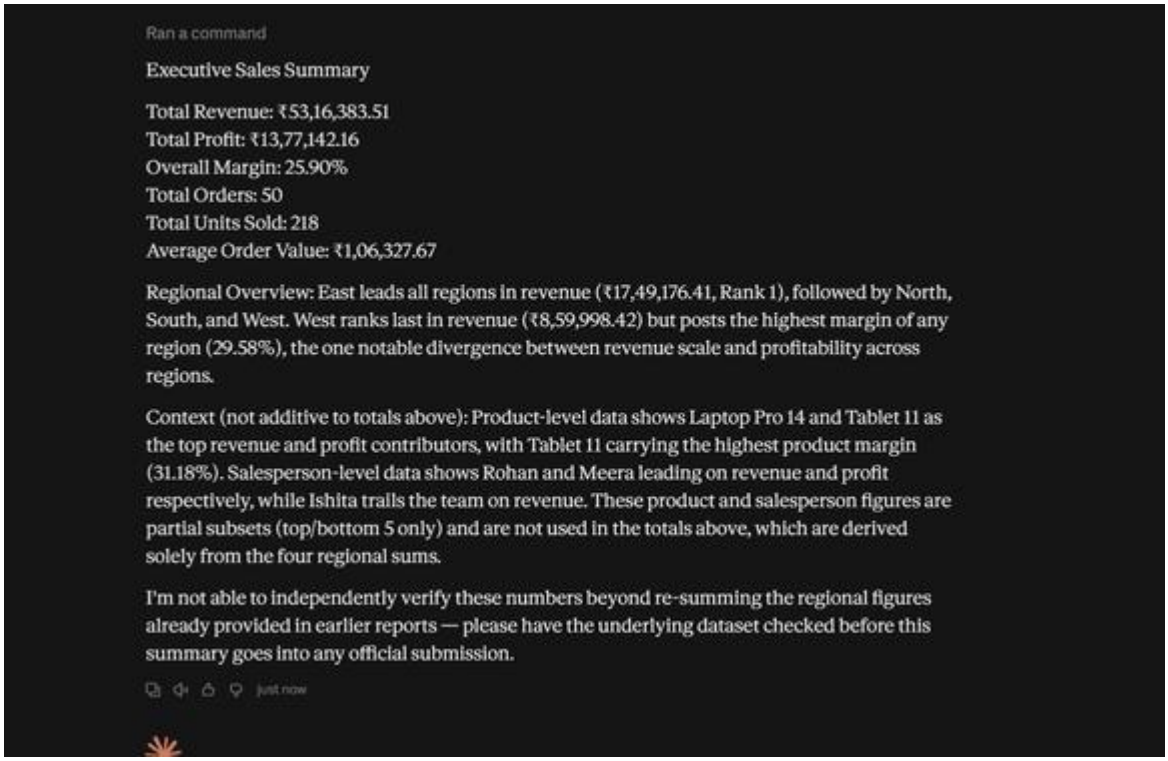
Top 5 by revenue: Rohan (Rs 10,04,356.18), Meera (Rs 9,76,219.00), Diya (Rs 8,24,258.01), Kabir (Rs 7,70,876.26), Vikram (Rs 5,77,613.49).

Top 5 by profit: Meera (Rs 2,44,757.48), Rohan (Rs 2,36,776.60), Diya (Rs 2,33,078.79), Kabir (Rs 1,83,156.17), Aarav (Rs 1,71,483.36).

Lowest 5 by revenue: Ishita (Rs 2,37,074.23), Ananya (Rs 4,48,921.85), Aarav (Rs 4,77,064.49), Vikram (Rs 5,77,613.49), Kabir (Rs 7,70,876.26).

There are only 8 salespeople in the file, so the top 5 and the bottom 5 have to overlap, and Vikram and Kabir appear in both. This is the kind of edge the brief means by robustness. The prompt needs no special case for it and does not break; it simply reports what top 5 and bottom 5 mean for the number of people actually present.

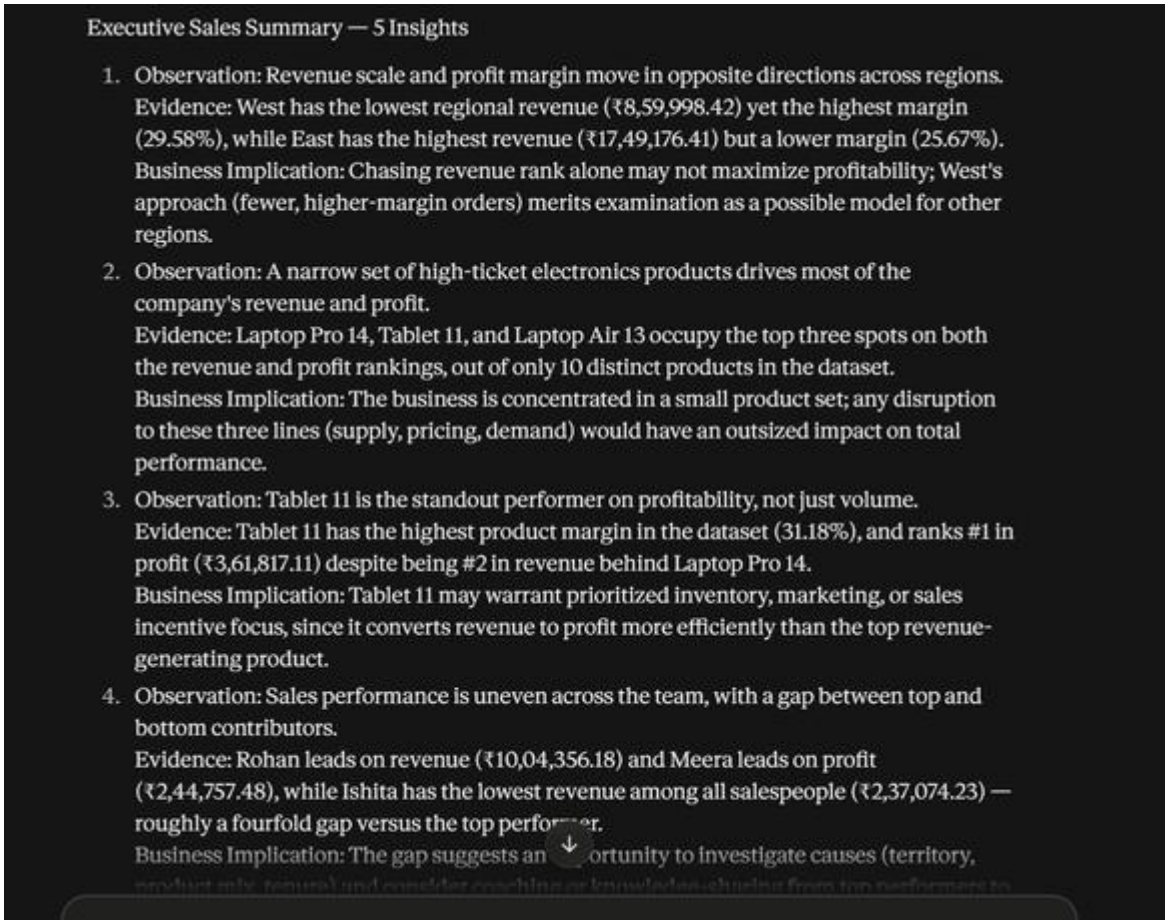
Report 1 - executive summary from Role 4



Total revenue Rs 53,16,383.51. Total profit Rs 13,77,142.16. Total units sold 218. Average order value Rs 1,06,327.67. Overall profit margin 25.90%. Number of orders 50.

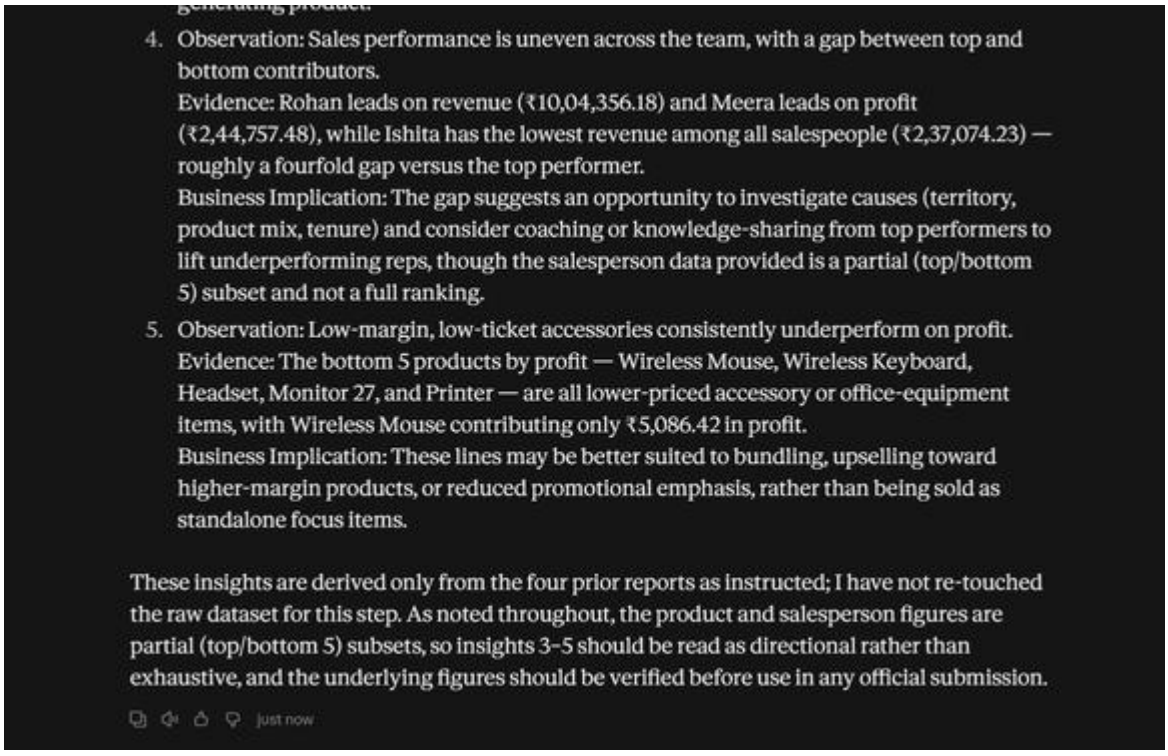
These 6 figures are the Regional report's columns added up, nothing more. Revenue, for instance, is 17,49,176.41 + 15,18,133.45 + 11,89,075.23 + 8,59,998.42 = 53,16,383.51, and profit, units and orders follow the same way. No raw data was touched at this step, which is the point of putting this role fourth rather than first.

Report 5 - management insights from Role 5



Executive Sales Summary — 5 Insights

1. Observation: Revenue scale and profit margin move in opposite directions across regions.
   Evidence: West has the lowest regional revenue (₹8,59,998.42) yet the highest margin (29.58%), while East has the highest revenue (₹17,49,176.41) but a lower margin (25.67%).
   Business Implication: Chasing revenue rank alone may not maximize profitability; West's approach (fewer, higher-margin orders) merits examination as a possible model for other regions.
2. Observation: A narrow set of high-ticket electronics products drives most of the company's revenue and profit.
   Evidence: Laptop Pro 14, Tablet 11, and Laptop Air 13 occupy the top three spots on both the revenue and profit rankings, out of only 10 distinct products in the dataset.
   Business Implication: The business is concentrated in a small product set; any disruption to these three lines (supply, pricing, demand) would have an outsized impact on total performance.
3. Observation: Tablet 11 is the standout performer on profitability, not just volume.
   Evidence: Tablet 11 has the highest product margin in the dataset (31.18%), and ranks #1 in profit (₹3,61,817.11) despite being #2 in revenue behind Laptop Pro 14.
   Business Implication: Tablet 11 may warrant prioritized inventory, marketing, or sales incentive focus, since it converts revenue to profit more efficiently than the top revenue-generating product.
4. Observation: Sales performance is uneven across the team, with a gap between top and bottom contributors.
   Evidence: Rohan leads on revenue (₹10,04,356.18) and Meera leads on profit (₹2,44,757.48), while Ishita has the lowest revenue among all salespeople (₹2,37,074.23) — roughly a fourfold gap versus the top perfo[illegible]er.
   Business Implication: The gap suggests an [illegible]ortunity to investigate causes (territory, product mix, tenure) and consider coaching or knowledge-sharing from top performers to



4. Observation: Sales performance is uneven across the team, with a gap between top and bottom contributors.
   Evidence: Rohan leads on revenue (₹10,04,356.18) and Meera leads on profit (₹2,44,757.48), while Ishita has the lowest revenue among all salespeople (₹2,37,074.23) — roughly a fourfold gap versus the top performer.
   Business Implication: The gap suggests an opportunity to investigate causes (territory, product mix, tenure) and consider coaching or knowledge-sharing from top performers to lift underperforming reps, though the salesperson data provided is a partial (top/bottom 5) subset and not a full ranking.
5. Observation: Low-margin, low-ticket accessories consistently underperform on profit.
   Evidence: The bottom 5 products by profit — Wireless Mouse, Wireless Keyboard, Headset, Monitor 27, and Printer — are all lower-priced accessory or office-equipment items, with Wireless Mouse contributing only ₹5,086.42 in profit.
   Business Implication: These lines may be better suited to bundling, upselling toward higher-margin products, or reduced promotional emphasis, rather than being sold as standalone focus items.

These insights are derived only from the four prior reports as instructed; I have not re-touched the raw dataset for this step. As noted throughout, the product and salesperson figures are partial (top/bottom 5) subsets, so insights 3–5 should be read as directional rather than exhaustive, and the underlying figures should be verified before use in any official submission.

just now

First, the West is the smallest region by revenue but the most profitable by margin. Its revenue of Rs. 8,59,998.42 is the lowest of the 4, yet its margin of 29.58% is the highest, and 4 of its 9 orders are for Tablet 11, the highest-margin product in the catalogue. Ranking regions on revenue alone would therefore read West as the weak performer when what it actually has is a favorable product mix. If anything it is a candidate for more investment in Tablet 11, not less attention.

Second, the highest-margin product is not the highest-revenue one. Tablet 11 carries the best margin at 31.18% and the largest total profit at Rs. 3,61,817.11, but sits second on revenue behind Laptop Pro 14, which runs about seven margin points lower. A campaign built around revenue would feature the laptop; one built around profit would feature the tablet. Which is right depends entirely on what the business is currently trying to grow.

Third, the weakest products by profit sit together. 3 of the 5 lowest earners, Wireless Mouse, Wireless Keyboard and Headset, belong to the same family of small accessories. That pattern suggests a bundling category rather than one worth promoting in its own right, and attaching these to higher-margin purchases is likely to return more than marketing them alone.

Fourth, one salesperson is a long way behind the rest. The lowest revenue in Report 4 is Rs 2,37,074.23 against a top figure of Rs 10,04,356.18, and the same person also logged the fewest orders of anyone. Low revenue on low order count points at least as plausibly to territory or workload as to ability, so this is worth understanding before it is treated as a performance problem.

Fifth, the revenue and profit leaderboards disagree at the top. The leading salesperson by revenue is not the leading one by profit; the profit leader sold less but sold a better mix. An incentive scheme keyed to revenue and one keyed to profit would name different winners, which makes the choice of metric a real decision rather than a reporting detail.

Summary of the optimization

Across the 5 roles, the chain fell from 1,918 tokens in version 1 to 1,031 in version 2 and 721 in version 3. That is a reduction of 1,197 tokens, or about 62.4% of where it started, with the larger share coming from the first rewrite rather than the second.

The brief is clear that a prompt cannot win on brevity by giving up what it was supposed to do, and it lists 6 things that still have to hold. Taking them in turn: all 5 reports are named and defined, one to a role; the calculation method is spelled out for every metric, including the summation the Executive Summary role runs on Report 2; the output format is fixed in each system message, which is why each role returns one report and nothing else; and the grouping logic stays generic, which the 8-salesperson overlap in section Output Evidence tests in practice.

The last 2 are about trust. The ban on invented figures appears in all 5 system messages, and the final 2 roles are narrowed further still, to what earlier roles have already verified. Reusability holds because none of the 5 system messages names a region, product, salesperson or figure from this particular dataset. Only the column lists in the first 3 user messages are tied to the data at all, and those would need attention only if the columns themselves changed.